

Validation and Test Metrics for Regression Models

This table summarizes the performance of all regression and ensemble models on the validation and test sets for the diabetes regression project. Lower MAE and MSE indicate better predictive accuracy, while higher R^2 indicates better fit.

Model	Val MAE	Val MSE	Val R^2	Test MAE	Test MSE	Test R^2
LinearRegression	38.22	2499.32	0.511	45.57	3139.37	0.447
DecisionTree	61.56	6207.92	-0.214	58.49	5195.27	0.085
RandomForest	39.25	2704.61	0.471	45.22	2904.35	0.488
GradientBoosting	42.35	2858.23	0.441	47.07	3281.49	0.422
KNN	41.46	2969.18	0.419	46.51	3471.28	0.389
XGBoost	41.84	3018.40	0.410	45.40	3318.62	0.416
Voting_Top3	38.82	2532.14	0.505	45.06	2953.74	0.480
Bayesian_Ensemble	42.46	3169.63	0.380	45.54	3069.84	0.459

Interpretation

Linear Regression achieved the strongest validation performance, with the lowest validation MSE of 2499.32 and the highest validation R^2 of 0.511.

On the test set, Random Forest performed best overall, with the lowest test MSE of 2904.35 and the highest test R^2 of 0.488.

The Voting_Top3 ensemble remained competitive, producing a test MSE of 2953.74 and a test R^2 of 0.480, which was slightly below Random Forest but stronger than several individual models. The Bayesian_Ensemble did not outperform the Voting approach on either validation or test metrics, indicating that the simpler ensemble strategy was more effective for this dataset.